Ctrl-F

Your AI-powered assistant for instant access to company information, policies, and resources

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1 Goals of Ctrl-F

1.1 Non-AI Goals

1. Provide employees with fast access to company information
2. Reduce time spent searching through manuals and documents
3. The system is available on request
4. Reduce workload on HR and IT helpdesk for repetitive questions
5. Improve onboarding of new employees

1.2 AI-Related Goals

1. Understand natural language questions - AI needed because employees ask questions in many different ways, not using exact keywords
2. Generate conversational answers - AI needed to provide direct answers instead of just returning document links
3. Handle follow-up questions - AI needed to maintain conversation context and understand references like "What about interns?"
4. Summarise long documents - AI needed to condense complex policies into short, digestible answers

2 Stakeholders

Users

- Employees
- New hires and interns
- Managers
- HR and IT staff

People Affected by the System

- HR department

- IT support staff
- Compliance officers
- Team leaders
- Employees who make decisions based on chatbot answers

Managers

- Course instructors
- Project supervisors
- Company knowledge owners
- Department managers

Regulators

- Data protection authorities (GDPR compliance)
- Internal compliance teams
- Labor and workplace regulators

3 Requirements

3.1 Functional Requirements (EARS)

Core Functionality

- **Req01:** When an employee submits a question, Ctrl-F shall accept the question in natural language.
- **Req02:** When an employee asks a question, Ctrl-F shall return an answer based on company knowledge sources.
- **Req03:** When requested information is found, Ctrl-F shall provide the answer in clear and concise language.
- **Req04:** When answering a question, Ctrl-F shall display the source document references used.

- **Req05**: If requested information cannot be found, Ctrl-F shall state that it does not know and suggest relevant contacts.

Context and Personalisation

- **Req06:** When an employee asks a follow-up question, Ctrl-F shall use prior conversation context to interpret it.
- **Req07:** While processing a request, Ctrl-F shall verify the employee's identity through authentication.
- **Req08:** When role-specific information is available, Ctrl-F shall tailor answers to the employee's department or role.

Feedback and Learning

- **Req09:** When an answer is presented, Ctrl-F shall allow the employee to provide helpful/not helpful feedback.
- **Req10:** When feedback is provided, Ctrl-F shall record it for later review.

3.2 Non-Functional Requirements (EARS)

External Interfaces

- **NfReq01:** When an employee accesses the system, Ctrl-F shall provide a web-based chat interface.
- **NfReq02:** When source references are displayed, Ctrl-F shall provide clickable links to original documents.
- **NfReq03:** When used on employee devices, Ctrl-F shall support desktop and mobile browsers.

Performance

- **NfReq04:** When a question is submitted, Ctrl-F shall return a response within 5 seconds for 90% of requests.

- **NfReq05:** While in normal operation, Ctrl-F shall support multiple concurrent users.

Attributes

- **NfReq06:** While handling information, Ctrl-F shall achieve 85% acceptable answer accuracy.
- **NfReq07:** When an answer is generated, Ctrl-F shall include source attribution for 95% of factual responses.
- **NfReq08:** While operating, Ctrl-F shall be available 90% of the time, excluding scheduled maintenance.

Constraints

- **NfReq09:** While processing personal data, Ctrl-F shall comply with privacy regulations (GDPR).
- **NfReq10:** When using documents, Ctrl-F shall use only approved and authorised data sources.
- **NfReq11:** Ctrl-F shall not use confidential conversations for model retraining without approval.

Safety

- **NfReq12:** If confidence is below 85%, Ctrl-F shall warn the employee and recommend human assistance.
- **NfReq13:** If the employee asks a question outside the company information scope, Ctrl-F shall inform them that it's outside scope.

3.3 AI-Related Requirements

AI-Related Functional Requirements

Req01, Req02, Req03 (Natural language understanding and answer generation)

Implementation: Use a pre-trained large language model (LLM) with Retrieval-Augmented Generation (RAG)

- Store company documents in a vector database
- Retrieve relevant passages when the user asks a question
- Pass passages to LLM to generate an answer

Why this approach: Training from scratch is not feasible; pre-trained models already understand language

Req04 (Source attribution)

- **Threshold:** 95% of answers should include sources
- **Reason:** Trust is critical in company settings; employees need to verify information for HR/compliance questions
- **Implementation:** System returns document references along with each answer from the retrieval step

Req06 (Conversation context)

- **Implementation:** Maintain conversation history and pass to LLM for each new question
- **Why needed:** Employees naturally ask follow-up questions like *what about managers*?

Req11 (Confidence-based warnings)

- **Threshold:** Warn if confidence < 85%
- **Reason:** Safer to admit uncertainty than give a wrong answer in an enterprise environment
- **Implementation:** Use retrieval relevance scores and prompt engineering to estimate confidence

AI-Related Non-Functional Requirements

Safety (NfReq06, NfReq07, NfReq12**)**

- 85% accuracy threshold is realistic for a course project while still being trustworthy
- Source attribution reduces harm from incorrect answers
- Low-confidence warnings prevent employees from acting on uncertain information

Security and Privacy (NfReq09, NfReq11, Req07**)**

- Must comply with GDPR for employee data
- Authentication ensures that only authorised users have access to restricted information
- No retraining on private conversations protects employee privacy

Explainability (Req04, NfReq07**)**

- The source display shows employees why a response was given
- Builds trust through transparency

Transparency and Trust (NfReq06, NfReq07, NfReq12, Req05**)**

- System admits when it doesn't know
- Shows confidence scores
- Always provides source references

Regulation (NfReq09, NfReq10**)**

- Must comply with privacy laws
- Only uses approved policy documents as sources
- Audit logs for compliance tracking

4 Use Case Descriptions

UC1 Ask Question and Receive Answer: Employee asks a question and receives an answer based on company knowledge. (Implemented)

UC2 Ask a follow-up question: Employee asks a follow-up question and the system uses prior conversation context. (Implemented)

UC3 Ask a role-specific question: User receives an answer that is tailored to the specific role or department. (Implemented)

UC4 Maintain Approved Knowledge Sources: A knowledge owner, department manager, or compliance officer manages the set of authorised company documents and sources that Ctrl-F is allowed to use for retrieval and answer generation. (Mostly implemented)

UC5 Review Feedback on Answers: HR staff, IT staff, knowledge owners, or managers review feedback submitted by employees about chatbot answers in order to identify weak answers, missing information, and opportunities to improve company knowledge sources and chatbot performance. (Implemented)

SUC1 Authenticate User: Verifies the identity of the employee before giving access to restricted information. (Simple demo authentication implemented)

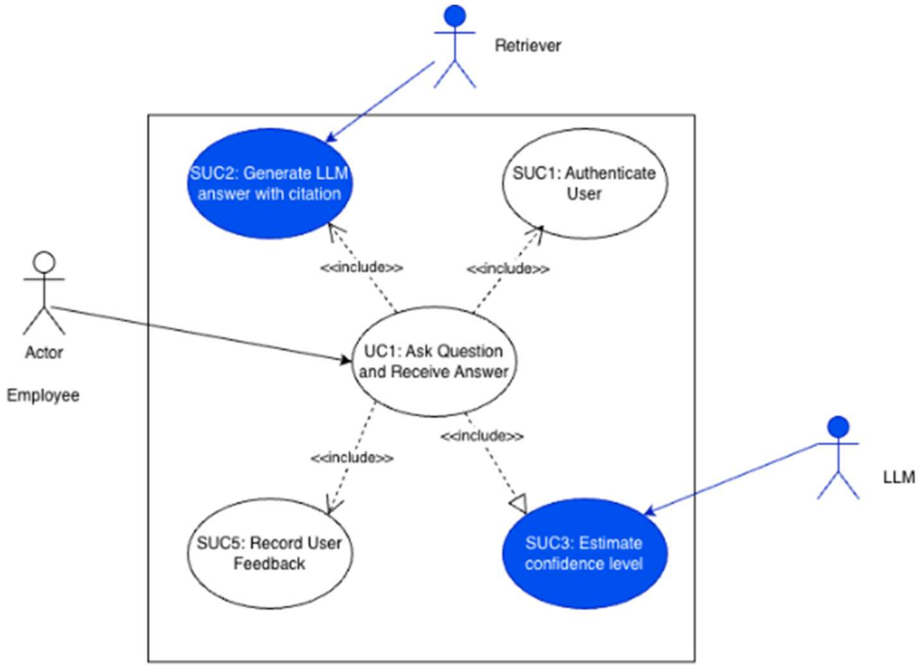
SUC2 Generate LLM answer with citation: Generates an answer based on the retrieved information and attaches source references. (Implemented)

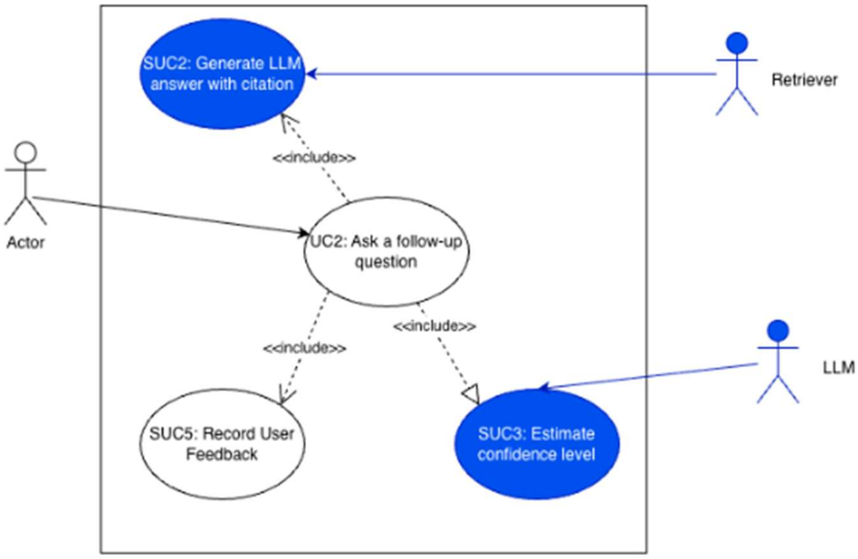
SUC3 Estimate confidence level: Calculates how confident the system is in the generated answer. (Implemented)

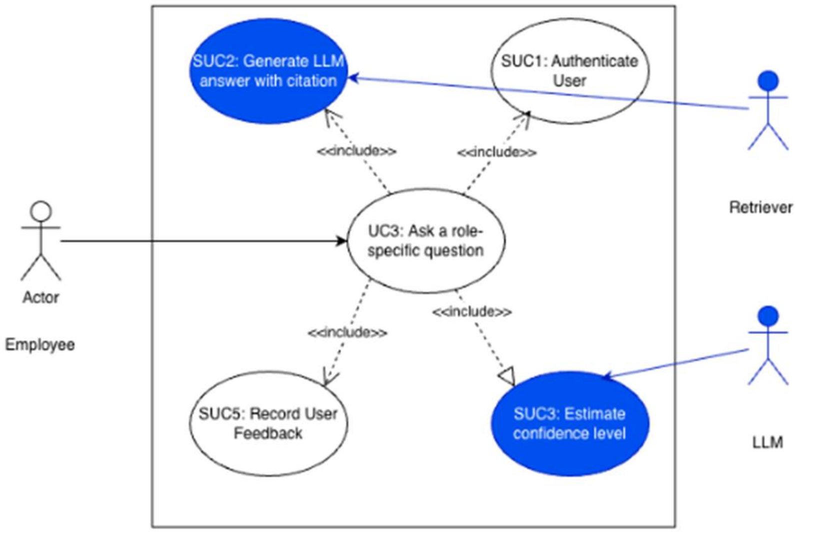
SUC4 Validate Source Authorization: Before a document or knowledge source is added, updated, or kept in the approved source set, Ctrl-F checks whether the source is authorised, valid, and allowed to be used for answer generation. (Partially implemented)

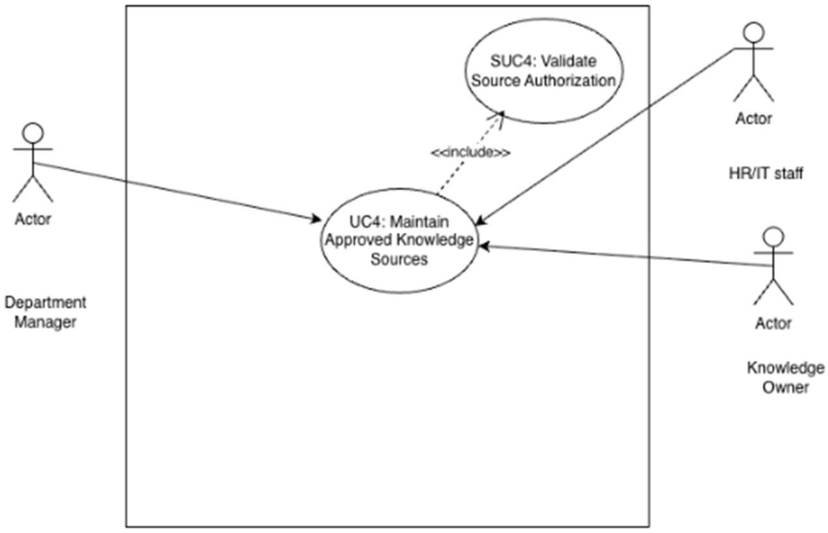
SUC5 Record User Feedback: Store helpful/not helpful feedback on an answer for later review. (Implemented)

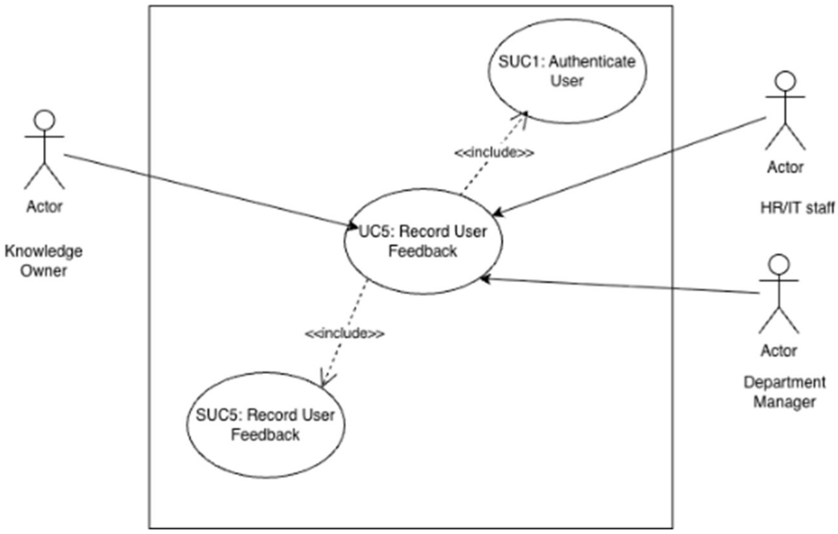
5 Use Case Diagram






SUC2: Generate LLM answer with citation
SUC1: Authenticate User
<<include>>
<<include>>
Retriever
Actor
Employee
UC3: Ask a role-specific question
<<include>>
<<include>>
SUC5: Record User Feedback
SUC3: Estimate confidence level
LLM

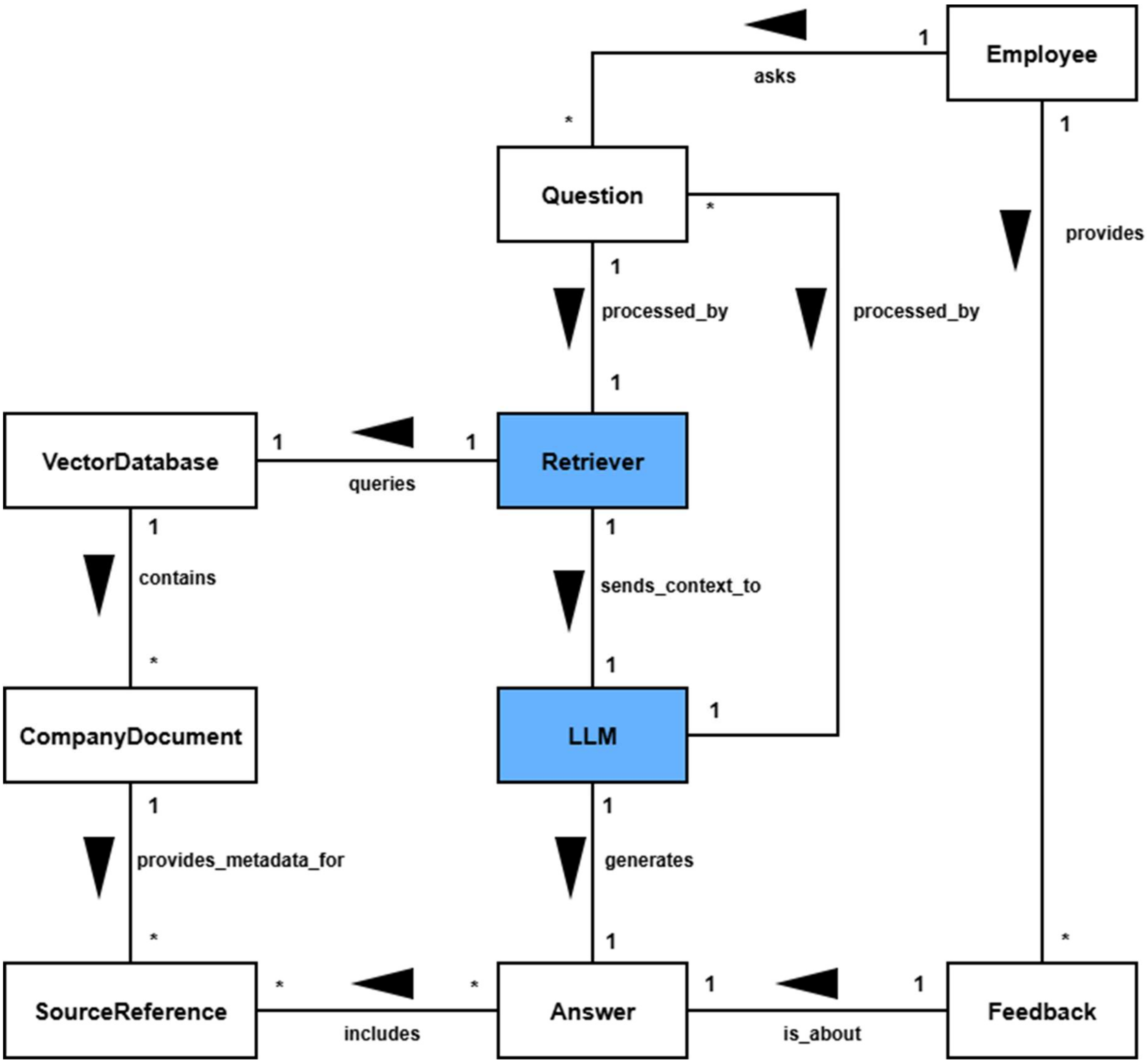

SUC4: Validate Source Authorization
Actor
HR/IT staff
<<include>>
Actor
Department Manager
UC4: Maintain Approved Knowledge Sources
Actor
Knowledge Owner


SUC1: Authenticate User
Actor
HR/IT staff
<<include>>
Actor
Knowledge Owner
UC5: Record User Feedback
Actor
Department Manager
<<include>>
SUC5: Record User Feedback

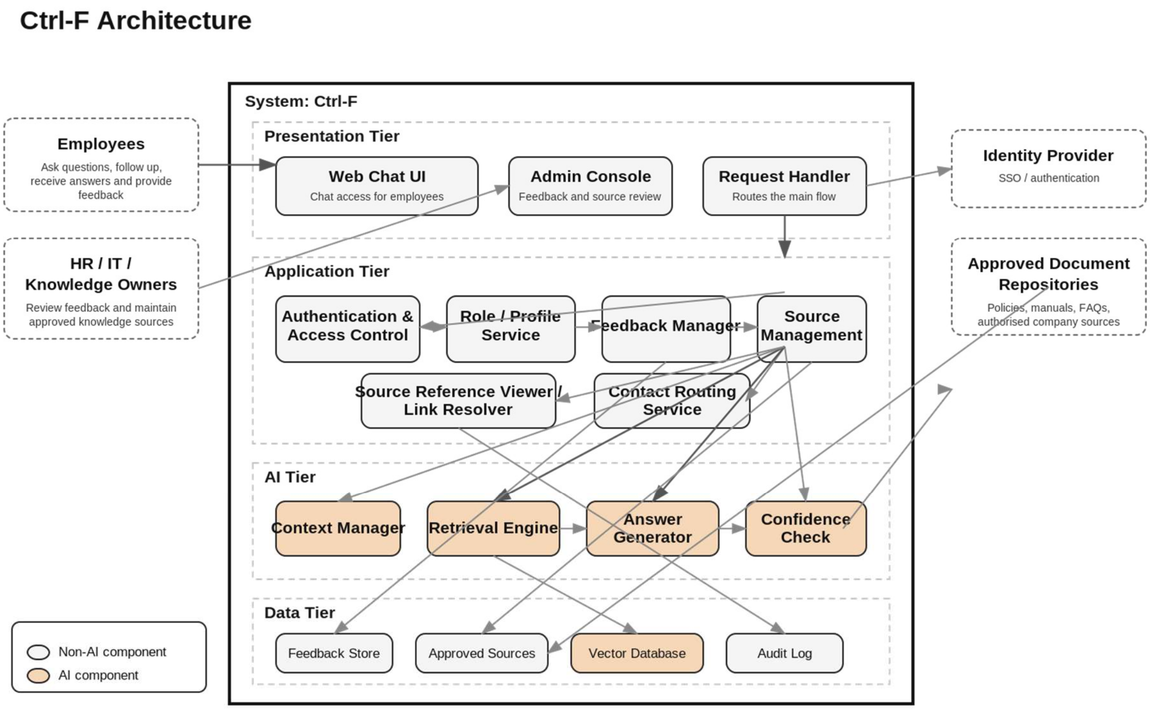
6 Traceability Matrix

Use cases	UC1	UC2	UC3	UC4	UC5	SUC1	SUC2	SUC3	SUC4	SUC5
Requirements										
Req1	TRUE	TRUE								
Req2	TRUE	TRUE	TRUE	TRUE			TRUE			
Req3	TRUE	TRUE	TRUE				TRUE			
Req4	TRUE	TRUE	TRUE	TRUE			TRUE			
Req5	TRUE	TRUE								
Req6		TRUE								
Req7	TRUE		TRUE			TRUE				
Req8			TRUE			TRUE				
Req9	TRUE	TRUE	TRUE		TRUE					TRUE
Req10					TRUE					TRUE
NfReq1	TRUE	TRUE	TRUE							
NfReq2	TRUE	TRUE	TRUE				TRUE			
NfReq3	TRUE	TRUE	TRUE							
NfReq4	TRUE	TRUE	TRUE				TRUE			
NfReq5	TRUE	TRUE	TRUE							
NfReq6	TRUE	TRUE	TRUE					TRUE		
NfReq7	TRUE	TRUE	TRUE				TRUE			
NfReq8	TRUE	TRUE	TRUE							
NfReq9			TRUE	TRUE		TRUE			TRUE	
NfReq10	TRUE		TRUE	TRUE			TRUE		TRUE	
NfReq11				TRUE						
NfReq12								TRUE		
NfReq13	TRUE	TRUE	TRUE							

7 Domain Model



8 Architecture Diagram



8.1 Components and relation to requirements

1. Web Chat UI

Provides the employee-facing chat interface where users ask questions, read answers, view sources, see warnings, and submit feedback. It is the main access point for employees on desktop and mobile browsers.

Related requirements: Req01, Req03, Req04, Req09, NfReq01, NfReq02, NfReq03, NfReq12

2. Request Handler

Coordinates the main request flow from the UI to the internal services. It routes incoming questions, handles errors, and returns the final response to the user.

Related requirements: Req01, Req02, Req05, NfReq04, NfReq05, NfReq08, NfReq13

3. Admin Console

Provides an internal interface for HR, IT, and knowledge owners to review answer feedback and maintain approved knowledge content. It supports operational oversight of the system.
Related requirements: Req10, NfReq09, NfReq10, NfReq11

4. Authentication & Access Control

Verifies the user's identity and ensures that only authorised employees can access protected information. It also helps enforce privacy and role-based restrictions.
Related requirements: Req07, Req08, NfReq09, NfReq10

5. Role/Profile Service

Stores or retrieves employee role, department, or employment status information. This allows the system to tailor answers to the correct user group.
Related requirements: Req08, Req07, NfReq09

6. Feedback Manager

Collects helpful/not helpful feedback from users and sends it to storage for later review. It supports continuous improvement of answers and knowledge coverage.
Related requirements: Req09, Req10, NfReq09, NfReq11

7. Source Management Service

Manages the list of approved documents and ensures that only authorised knowledge sources are made available to the system. It supports source governance and compliance.
Related requirements: NfReq10, NfReq09, Req02, Req04

8. Context Manager

Maintains relevant conversation history so the system can understand follow-up questions correctly. It helps preserve meaning across multiple turns in the same chat.
Related requirements: Req06, NfReq11

9. Retrieval Engine

Searches the approved company knowledge base for passages relevant to the employee’s question. It is the retrieval part of the RAG approach.
Related requirements: Req02, Req04, Req05, NfReq04, NfReq06, NfReq07, NfReq10

10. Answer Generator

Generates a clear natural-language answer based on the retrieved passages and user context. It is responsible for concise conversational responses instead of just returning links.
Related requirements: Req01, Req02, Req03, Req04, Req06, Req08, NfReq06, NfReq07

11. Confidence Check

Estimates how reliable the generated answer is before it is shown to the user. If the confidence is too low, it triggers a warning and recommends human help.
Related requirements: Req05, NfReq06, NfReq12

12. Feedback Store

Stores submitted feedback together with the related interaction data. This makes later review and quality analysis possible.
Related requirements: Req10, NfReq09, NfReq11

13. Approved Sources

Stores the authorised set of company documents and repositories that Ctrl-F is allowed to use. It ensures that retrieval is limited to approved knowledge only.
Related requirements: NfReq10, NfReq09, Req02, Req04

14. Vector Database

Stores document embeddings and supports fast semantic retrieval of relevant passages. It is a core data component of the RAG pipeline.
Related requirements: Req02, NfReq04, NfReq05, NfReq06, NfReq10

15. Audit Log

Records important system events such as access, source usage, and administrative changes. It supports traceability, compliance, and investigation of system behaviour.
Related requirements: Req04, NfReq09, NfReq11

9 Design Questions

DQ1: How do we ensure that generated answers stay grounded in company documents instead of hallucinating unsupported information?
A1: If the Vector DB returns zero highly-relevant company documents for a prompt, the system will automatically assume the question is out of scope and trigger a hardcoded fallback response rather than letting the LLM hallucinate.

DQ2: How should conversation history be managed and bounded to prevent context window overflow in the LLM?
A2: The Context Manager will maintain a sliding window of the last N turns of conversation, where N is determined based on the LLM's context window size. Each message will be stored with a token count. When the accumulated token count approaches the model limit, the oldest turns are removed. A summary of removed turns can optionally be prepended.

DQ3: How do we restrict answers so users only receive information permitted for their role or department?
A3: The system will authenticate the user through the external identity provider and then apply role-based access filters before retrieval. Restricted documents will never be passed to the retriever or LLM unless the user is authorized to access them.

DQ4: How should the system handle a situation where two approved documents contradict each other?
A4: The system should detect conflicting passages and avoid presenting a single definitive answer without warning. The generated answer should present both perspectives and note that the documents disagree, along with references to both sources. Additionally, the system refers to the responsible human department.

DQ5: Will the system accurately understand questions from users who are not native English speakers or who use poor grammar?
A5: Because we are using a modern, pre-trained LLM (which is trained on diverse, multilingual internet data), it has a high natural tolerance for typos, poor grammar, and varying phrasing. We will test edge cases to ensure the embedding model accurately maps poorly phrased questions to the correct policies.

DQ6: How do we define and measure correctness for our chatbot against a base rate?
A6: Correctness is defined as providing factually accurate answers supported by an internal document. We will measure this by creating a dataset of 100 historical HR questions and comparing the chatbot's acceptable answer rate (target 85%) against the current human-HR first-contact resolution rate.

DQ7: How should the system estimate confidence well enough to decide when to warn the user or escalate to human support?
A7: Confidence will be estimated from several signals combined: retrieval score quality, number of strong supporting passages, citation coverage, and model self-check rules. If the weighted combination of these scores is below the 85% threshold, the answer will include a warning or be escalated.

DQ8: How can we measure whether the chatbot is actually helping employees instead of only measuring technical accuracy?
A8: We will combine technical metrics such as retrieval quality and latency with user-centered metrics such as helpful feedback rate, escalation rate, and repeated-

question frequency. This gives a better view of whether the system supports real employee tasks.

DQ9: How should feedback be collected and used without violating the requirement that private conversations are not retrained on without approval?
A9: Feedback will be stored separately from training data and used first for analytics, evaluation, and manual review. Conversations will not be used for retraining unless explicit approval is given, which satisfies the privacy constraint.

DQ10: How should the user interface present answers with low certainty to the user without undermining their trust in the system?
A10: If the confidence score falls below 85%, the UI will explicitly display a warning banner (for example "I am not completely sure, but here is what I found") and automatically provide a button to "Escalate to Human HR/IT support".